Graphs of Sine and Cosine Functions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Amplitude, period, and midline

For the function $y = 3\sin(2x)$, state:

a the amplitude **b** the period **c** the range

2 [Foundational] • Target: Amplitude, period, and midline

For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

a the amplitude **b** the period

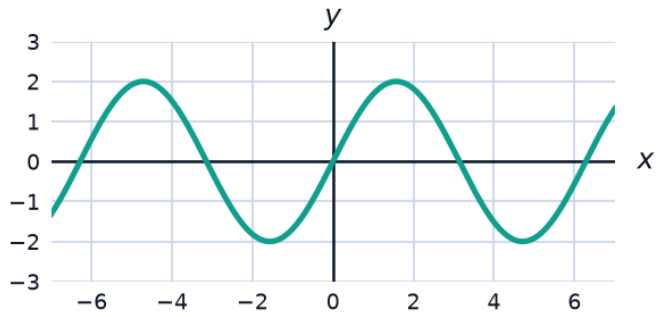
c the midline **d** the range

3 [Foundational] • Target: Amplitude, period, and midline

For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

4 [Foundational] • Target: Reading and writing equations from graphs

The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.



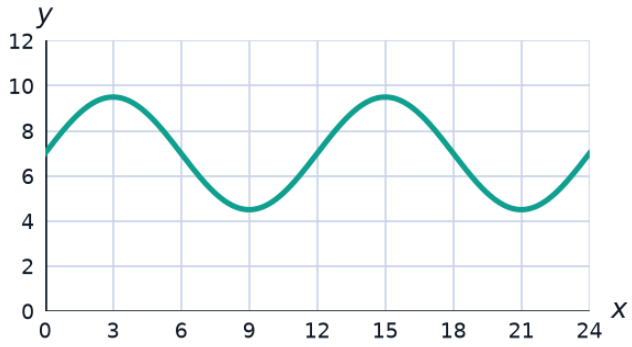
5 [Foundational] • Target: Reading and writing equations from graphs

Write an equation of a cosine function with amplitude 4, period π, reflected across the x-axis, with no phase or vertical shift.

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6 [Foundational] • Target: Reading and writing equations from graphs

The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.



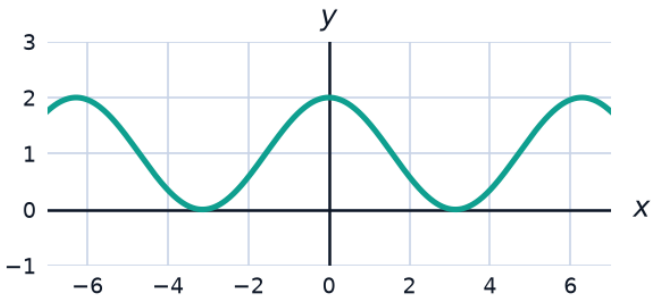
a State the maximum and minimum depths.

b State the period of the tide cycle.

c Find the depth at $t = 3$.

7 [Foundational] • Target: Graphing with transformations

The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.

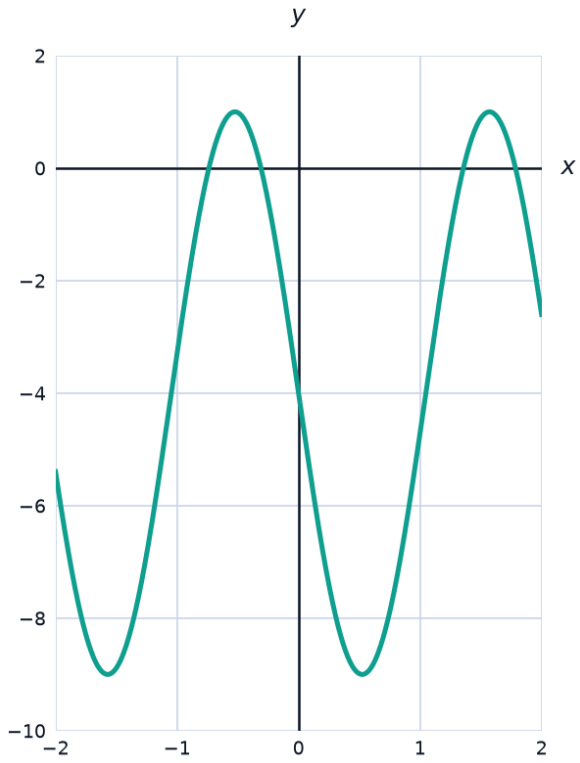


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Graphing with transformations

Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.



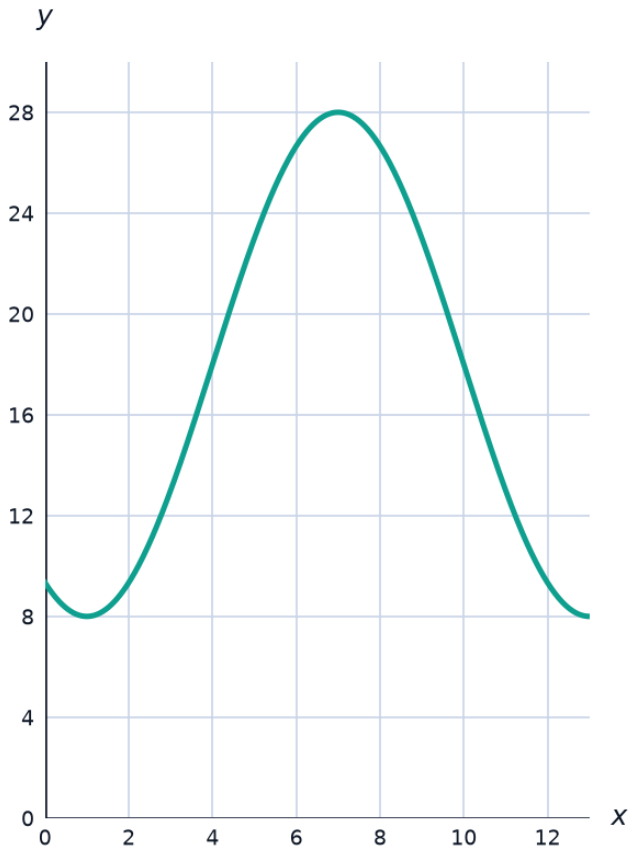
9 [Intermediate] • Target: Graphing with transformations

State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

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10 **[Intermediate]** • **Target: Modeling periodic phenomena**

The average monthly temperature in a city varies between $8°\text{C}$ (in January, month 1) and $28°\text{C}$ (in July, month 7), following a sinusoidal pattern with a 12-month period.



a Find the amplitude and midline of the model.

b Write a cosine model $T(m)$ for the temperature in month m, using the fact that the minimum occurs at $m = 1$.

c Use your model to estimate the temperature in month 4 (April).

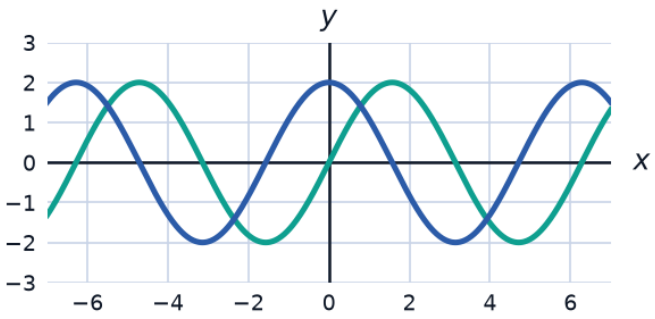
11 **[Intermediate]** • **Target: Comparing sine and cosine graphs**

Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.

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12 [Intermediate] • Target: Comparing sine and cosine graphs

Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.



13 [Intermediate] • Target: Finding zeros and extrema from an equation

Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.

14 [Intermediate] • Target: Finding zeros and extrema from an equation

Find the x-value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

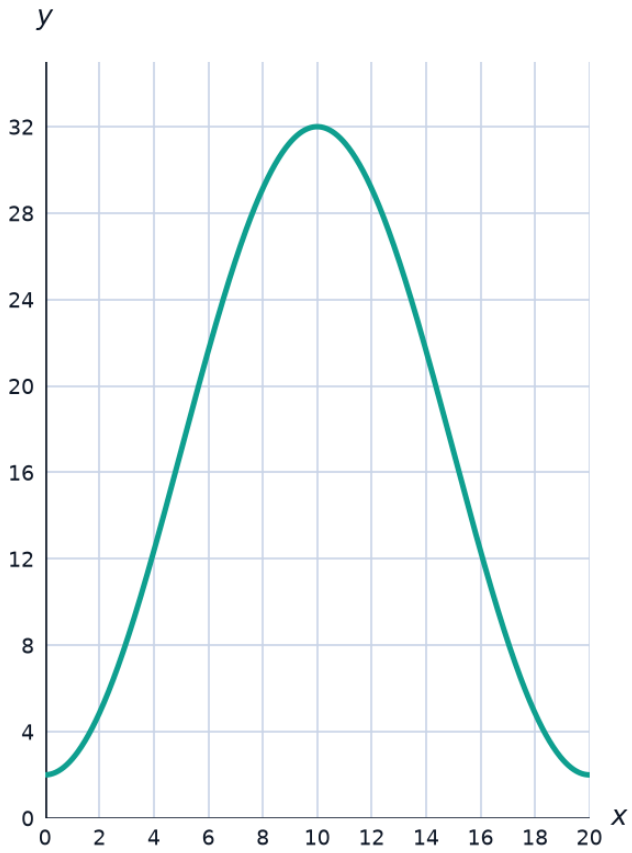
15 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 7.02 Logarithms and Their Properties

Solve $2\sin(x) + 1 = 0$ for $x \in [0, 2\pi]$.

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16 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A Ferris wheel's height above ground is $h(t) = 15\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) + 17$ meters. Find the first time $t > 0$ (minutes) when the rider is at height 17 m and rising.



17 [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the period and phase shift of $y = \tan(3x - 1)$.

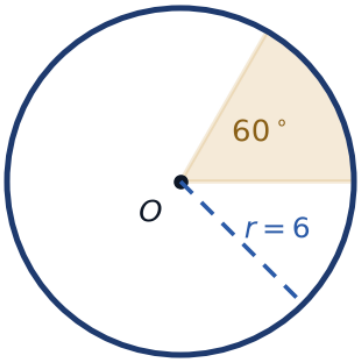
18 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

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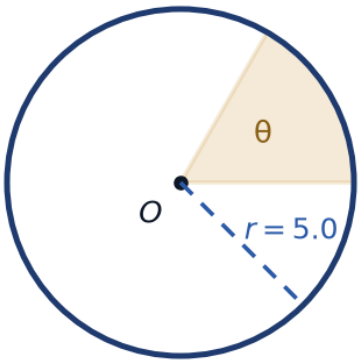
19 [Challenging] • Target: Amplitude and Vertical Shift • Cross-Lesson: 8.02 Graphing Rational Functions

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



20 [Challenging] • Target: Period and Frequency • Cross-Lesson: 8.03 Introduction to Limits

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Amplitude and Vertical Shift

Find the amplitude and midline equation of $y = -4\cos(x) + 3$.

a. Amplitude $= 4$, Midline: $y = 3$

b. Amplitude $= -4$, Midline: $y = 3$

c. Amplitude $= 4$, Midline: $y = -3$

d. Amplitude $= 8$, Midline: $y = 3$

22 [Foundational] • Target: Amplitude and Vertical Shift

What are the maximum and minimum values of $f(x) = 5\sin(x) - 2$?

a. Maximum $= 3$, Minimum $= -7$

b. Maximum $= 5$, Minimum $= -5$

c. Maximum $= 7$, Minimum $= -3$

d. Maximum $= 3$, Minimum $= -3$

23 [Foundational] • Target: Amplitude and Vertical Shift

A wave oscillates between a maximum of 12 and a minimum of 4. Find its amplitude.

a. 4

b. 8

c. 6

d. 16

24 [Foundational] • Target: Period and Frequency

Find the period of the function $y = 3\sin(4x)$.

a. $\frac{\pi}{2}$

b. π

c. 2π

d. $\frac{\pi}{4}$

25 [Foundational] • Target: Period and Frequency

Find the frequency (cycles per unit of x) of $y = \cos\left(\frac{\pi}{3}x\right)$.

a. 1/6

b. 6

c. $\pi/3$

d. 3

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26 [Foundational] • Target: Period and Frequency

Write a sine function with amplitude 3 and period 4π.

a. $y = 3\sin\left(\frac{1}{2}x\right)$

b. $y = 3\sin(4x)$

c. $y = 3\sin(2x)$

d. $y = 4\sin\left(\frac{1}{3}x\right)$

27 [Foundational] • Target: Phase Shift (Horizontal Shift)

Find the phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

a. $\frac{\pi}{6}$ to the right

b. $\frac{\pi}{2}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{3\pi}{2}$ to the right

28 [Foundational] • Target: Phase Shift (Horizontal Shift)

At what value of x does the first cycle start for $y = \cos(2x + \pi)$?

a. $-\frac{\pi}{2}$

b. π

c. $-\pi$

d. $\frac{\pi}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Phase Shift (Horizontal Shift)

Which transformation shifts the graph of $y = \cos x$ to become identical to $y = \sin x$?

a. Shift right by $\frac{\pi}{2}$

b. Shift left by $\frac{\pi}{2}$

c. Reflect across the x-axis

d. Shift up by 1

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30 [Intermediate] • Target: Modeling Periodic Phenomena

The temperature in a greenhouse follows $T(t) = 15\sin\left(\frac{\pi}{12}(t-6)\right) + 65$, where t is hours after midnight. What is the maximum temperature?

a. 80°F

b. 65°F

c. 75°F

d. 95°F

31 [Intermediate] • Target: Modeling Periodic Phenomena

In the greenhouse model above, at what time of day (t) does the maximum temperature occur?

a. $t = 12$ (12:00 PM / Noon)

b. $t = 6$ (6:00 AM)

c. $t = 18$ (6:00 PM)

d. $t = 24$ (Midnight)

32 [Intermediate] • Target: Modeling Periodic Phenomena

A Ferris wheel of radius 30 ft has its lowest point 5 ft off the ground and completes one rotation every 60 seconds. What is the midline height?

a. 35 ft

b. 30 ft

c. 65 ft

d. 40 ft

33 [Intermediate] • Target: Reading and writing equations from graphs

Find the period and phase shift of $y = \tan(2x - 1)$.

a. Period $\frac{1}{2}\pi$, phase shift $\frac{1}{2}$ units right

b. Period $\frac{1}{2}\pi$, phase shift $\frac{1}{2}$ units left

c. Period $\frac{1}{2}\pi$, phase shift 0

d. Period π, phase shift $\frac{1}{2}$ units right

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34 [Intermediate] • Target: Graphing with transformations

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. Period 2π, phase shift 2 units right

35 [Intermediate] • Target: Modeling periodic phenomena

Find the period and phase shift of $y = \tan(1/2x - 2)$.

a. Period 2π, phase shift 0

b. Period π, phase shift 4 units right

c. Period 2π, phase shift 4 units right

d. Period 2π, phase shift 4 units left

36 [Intermediate] • Target: Comparing sine and cosine graphs

Find the period and phase shift of $y = \tan(x)$.

a. Period 2π, phase shift 0

b. Period π, phase shift 0

c. Period π, phase shift 0 units left

d. Period π, phase shift 0 units right

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37 [Intermediate] • Target: Finding zeros and extrema from an equation

Find the period and phase shift of $y = \tan(x - 1)$.

a. Period π, phase shift 1 units right

b. Period π, phase shift 1 units left

c. Period π, phase shift 0

d. Period 2π, phase shift 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 8.01 Rational Functions and Asymptotes

What is the amplitude and period of $y = 5\sin(3x) - 3$?

a. Amplitude -3, period $\frac{2}{3}\pi$

b. Amplitude 5, period $\frac{2}{3}\pi$

c. Amplitude 5, period $\frac{3}{2}\pi$

d. Amplitude 5, period 2π

39 [Challenging] • Target: Amplitude and Vertical Shift • Cross-Lesson: 8.02 Graphing Rational Functions

What is the amplitude and period of $y = 3\sin(3x) + 5$?

a. Amplitude 3, period $\frac{3}{2}\pi$

b. Amplitude 5, period $\frac{2}{3}\pi$

c. Amplitude 3, period 2π

d. Amplitude 3, period $\frac{2}{3}\pi$

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40 [Challenging] • Target: Period and Frequency • Cross-Lesson: 8.03 Introduction to Limits

What is the amplitude and period of $y = 1\sin(3x) + 2$?

a. Amplitude 1, period 2π

b. Amplitude 2, period $\frac{2}{3}\pi$

c. Amplitude 1, period $\frac{2}{3}\pi$

d. Amplitude 1, period $\frac{3}{2}\pi$

41 [Challenging] • Target: Phase Shift (Horizontal Shift) • Cross-Lesson: 9.01 Arithmetic Sequences and Series

What is the amplitude and period of $y = 1\sin(3x) - 3$?

a. Amplitude 1, period 2π

b. Amplitude -3, period $\frac{2}{3}\pi$

c. Amplitude 1, period $\frac{3}{2}\pi$

d. Amplitude 1, period $\frac{2}{3}\pi$

42 [Challenging] • Target: Modeling Periodic Phenomena • Cross-Lesson: 9.02 Geometric Sequences and Series

What is the amplitude and period of $y = 1\sin(3x)$?

a. Amplitude 1, period 2π

b. Amplitude 1, period $\frac{2}{3}\pi$

c. Amplitude 1, period $\frac{3}{2}\pi$

d. Amplitude 0, period $\frac{2}{3}\pi$

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43 [Challenging] • Target: Amplitude, period, and midline • Cross-Lesson: 9.03 The Binomial Theorem

What is the amplitude and period of $y = 3\sin(3x) - 3$?

a. Amplitude 3, period $\frac{3}{2}\pi$

b. Amplitude 3, period $\frac{2}{3}\pi$

c. Amplitude -3, period $\frac{2}{3}\pi$

d. Amplitude 3, period 2π

44 [Challenging] • Target: Reading and writing equations from graphs • Cross-Lesson: 1.01 Functions and Function Notation

What is the amplitude and period of $y = 2\sin(3x) - 3$?

a. Amplitude 2, period $\frac{2}{3}\pi$

b. Amplitude -3, period $\frac{2}{3}\pi$

c. Amplitude 2, period $\frac{3}{2}\pi$

d. Amplitude 2, period 2π

45 [Challenging] • Target: Graphing with transformations • Cross-Lesson: 1.02 Domain and Range

What is the amplitude and period of $y = 2\sin(3x) + 2$?

a. Amplitude 3, period $\frac{2}{3}\pi$

b. Amplitude 2, period 2π

c. Amplitude 2, period $\frac{2}{3}\pi$

d. Amplitude 2, period $\frac{3}{2}\pi$

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